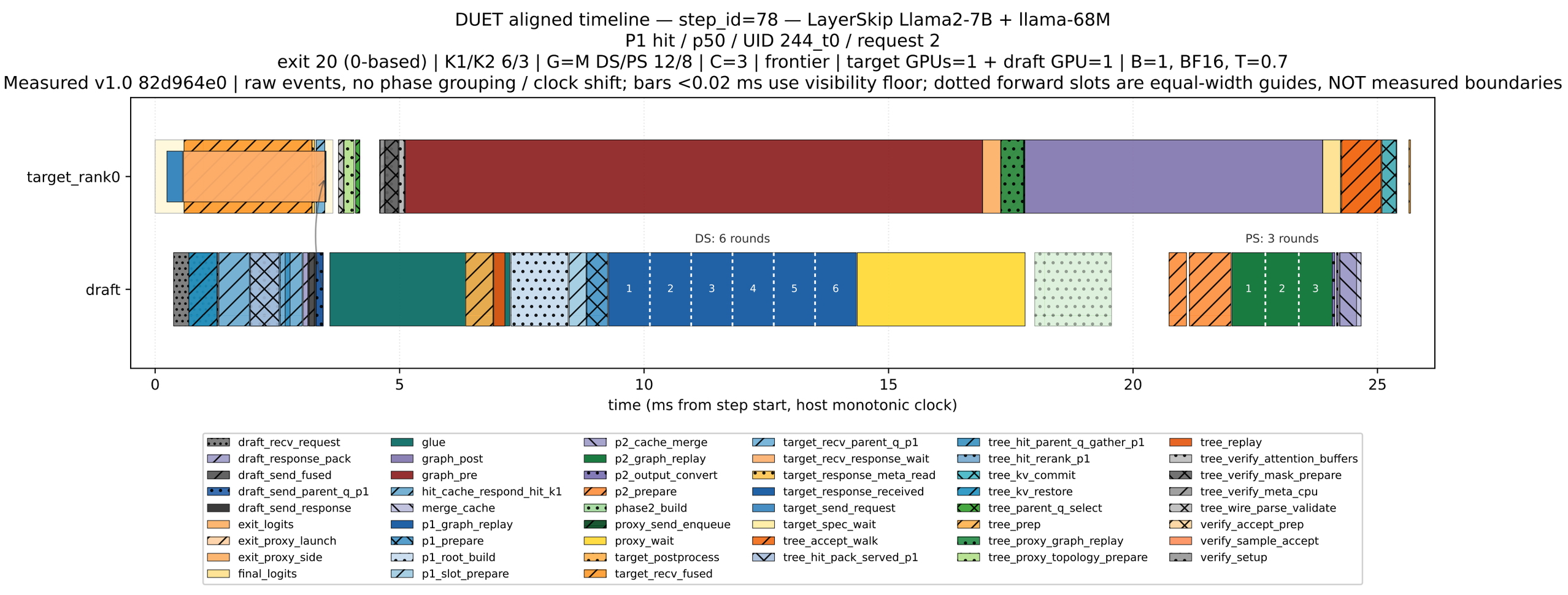
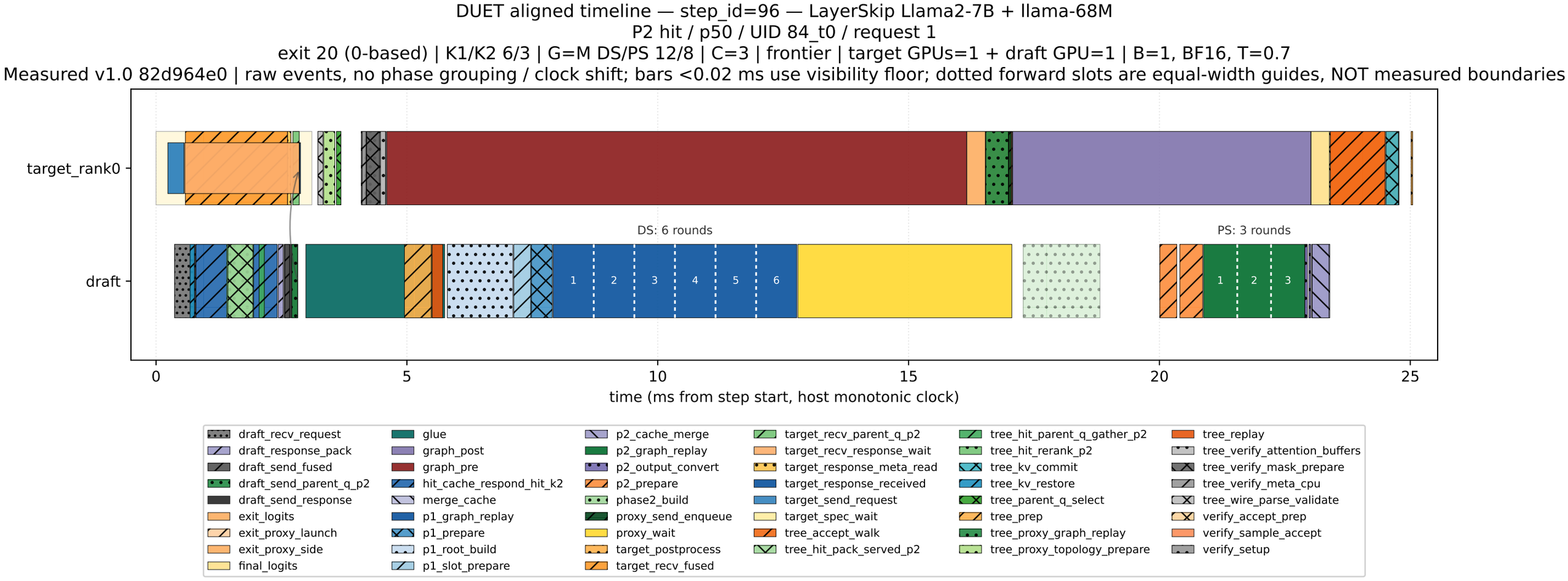
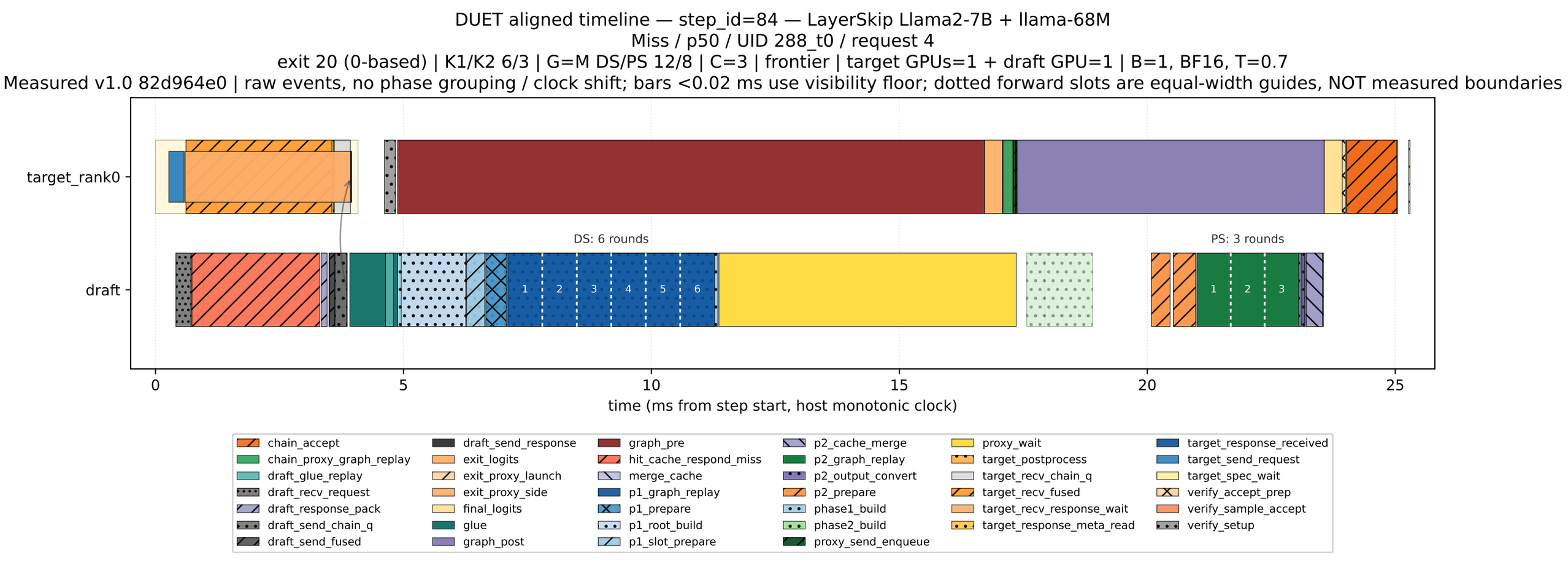
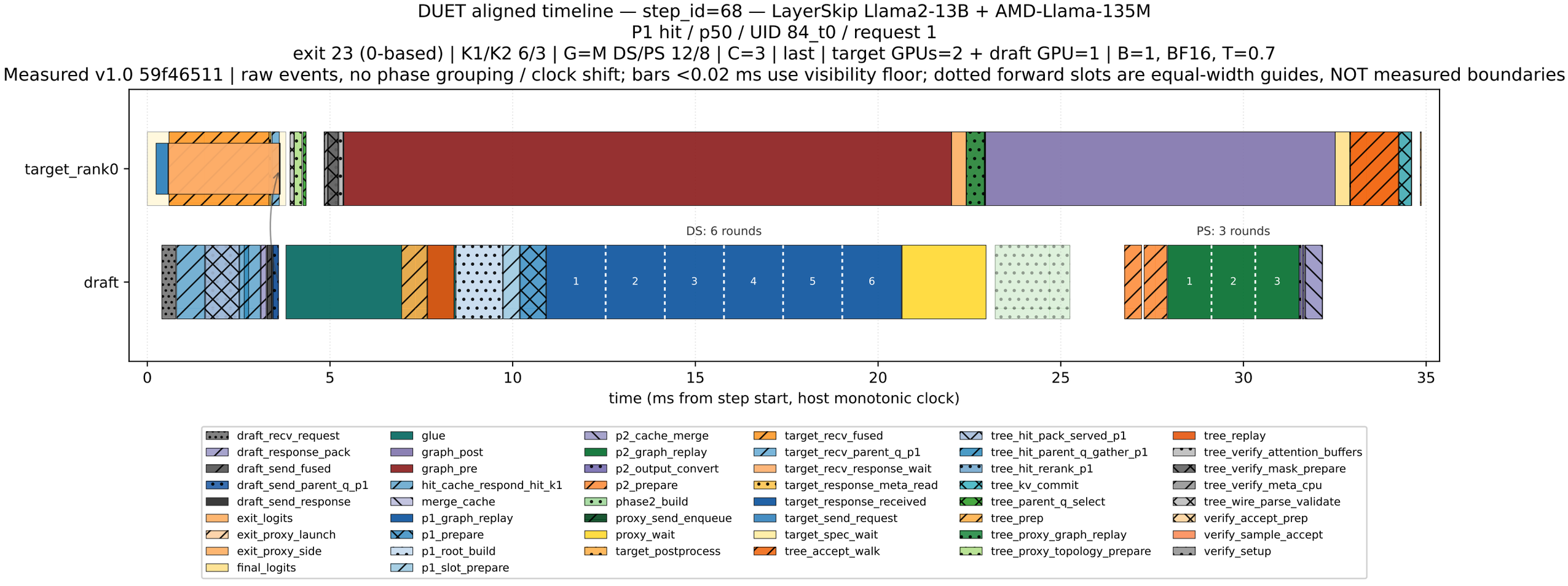
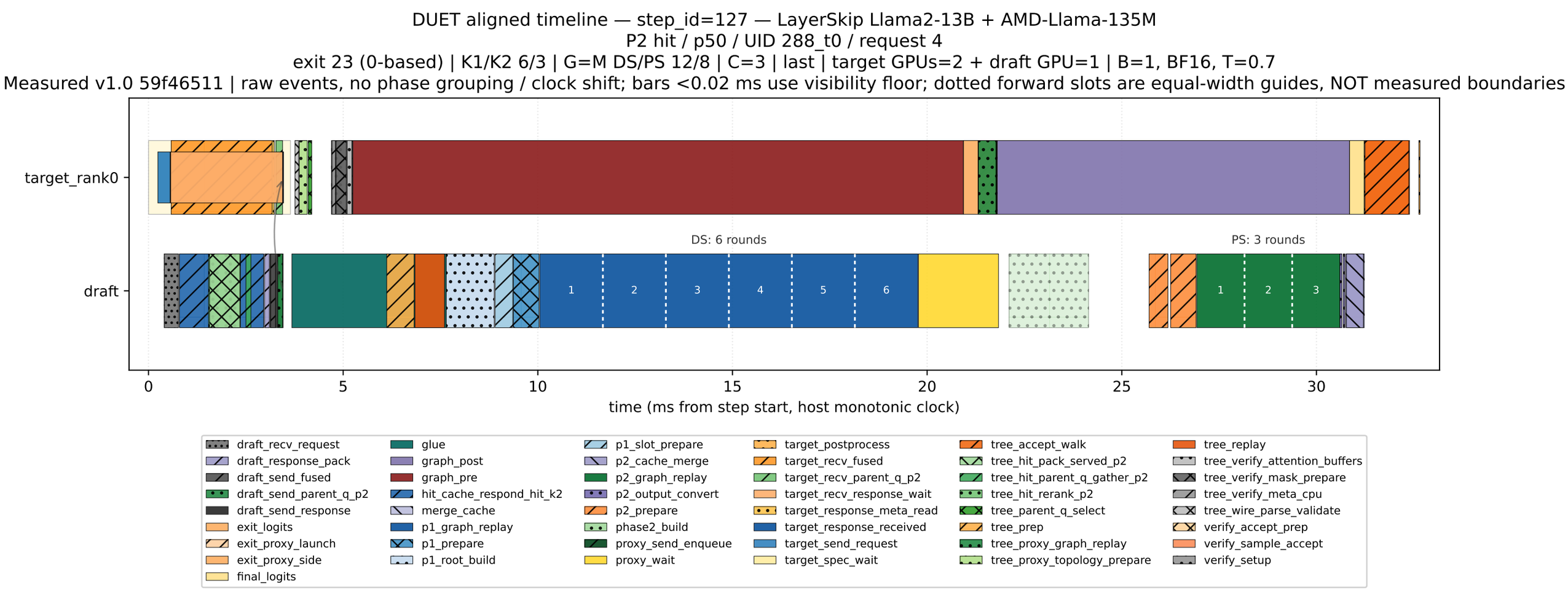
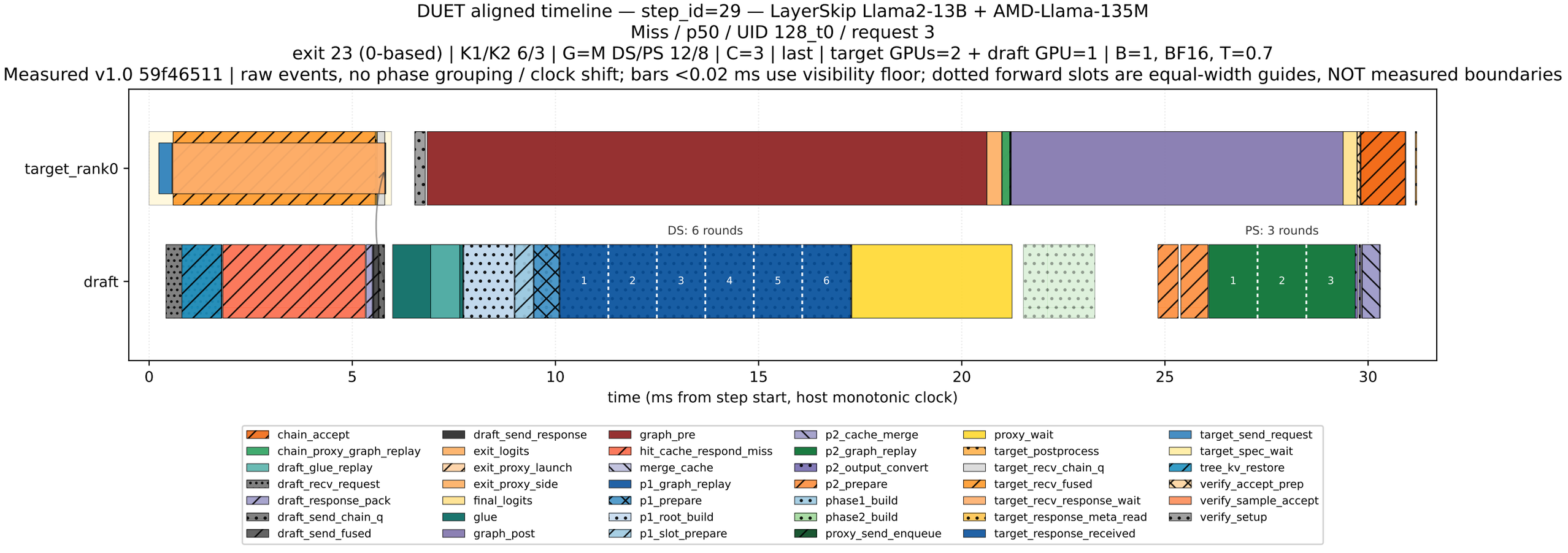

DUET aligned timeline — step_id=78 — LayerSkip Llama2-7B + llama-68M
P1 hit / p50 / UID 244_t0 / request 2
exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7
Measured v1.0 82d964e0 | raw events, no phase grouping / clock shift; bars <0.02 ms use visibility floor; dotted forward slots are equal-width guides, NOT measured boundaries
target_rank0
draft
DS: 6 rounds
PS: 3 rounds
1
2
3
4
5
6
1
2
3
0
5
10
15
20
25
time (ms from step start, host monotonic clock)
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p1
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
p2_prepare
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p1
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_parent_q_gather_p1
tree_hit_rerank_p1
tree_kv_commit
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

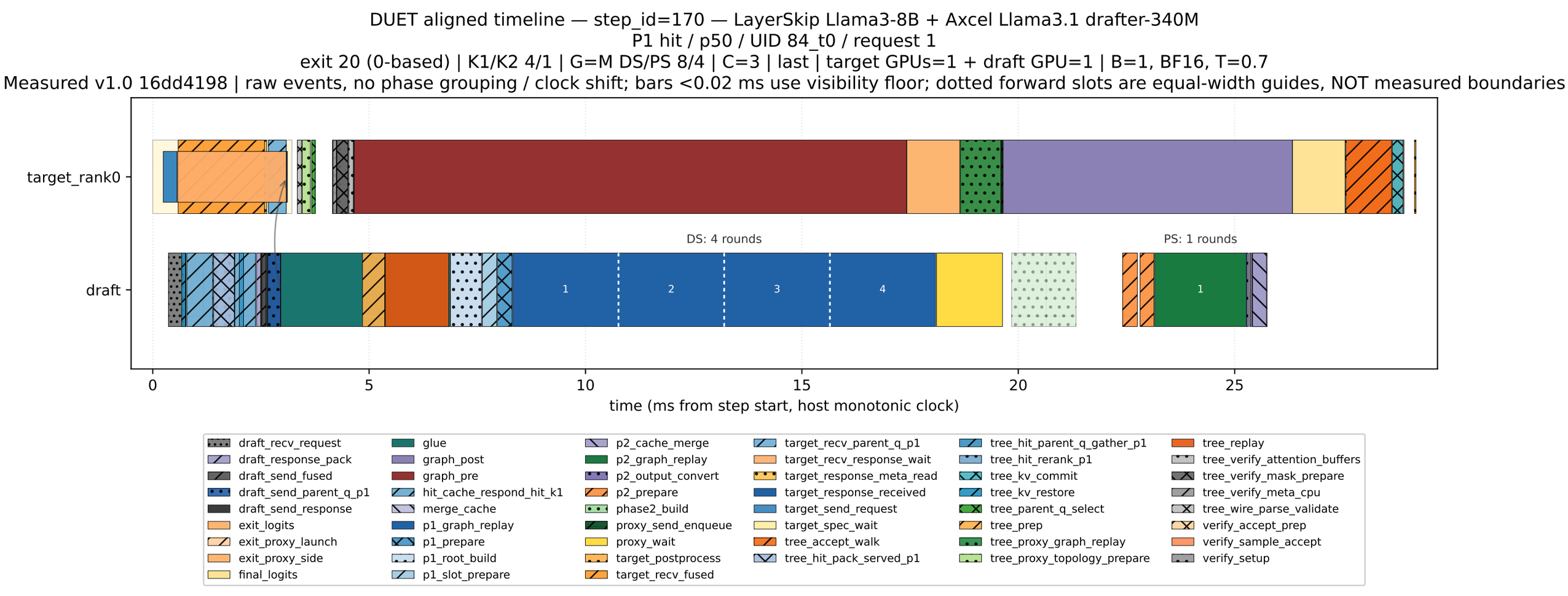

DUET aligned timeline — step_id=96 — LayerSkip Llama2-7B + llama-68M
P2 hit / p50 / UID 84_t0 / request 1
exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7
Measured v1.0 82d964e0 | raw events, no phase grouping / clock shift; bars <0.02 ms use visibility floor; dotted forward slots are equal-width guides, NOT measured boundaries
target_rank0
draft
DS: 6 rounds
PS: 3 rounds
1
2
3
4
5
6
1
2
3
0
5
10
15
20
25
time (ms from step start, host monotonic clock)
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k2
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
p2_prepare
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p2
tree_hit_rerank_p2
tree_kv_commit
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

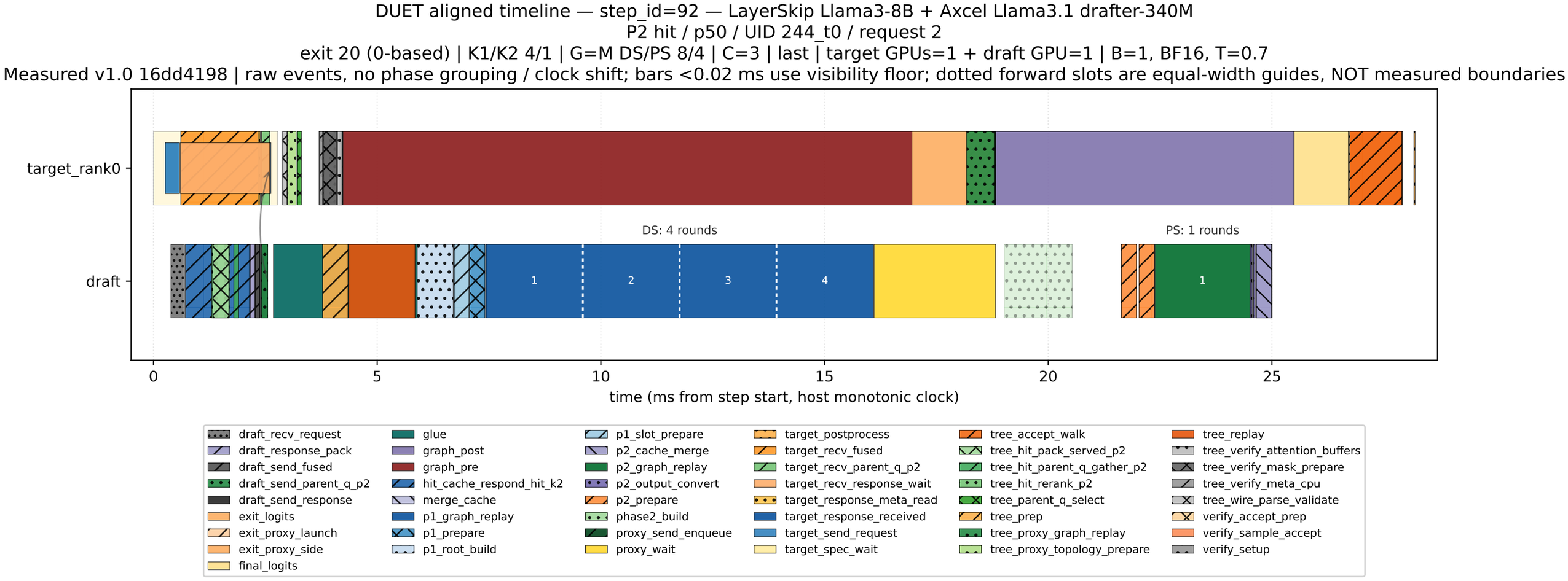

DUET aligned timeline — step_id=84 — LayerSkip Llama2-7B + llama-68M
Miss / p50 / UID 288_t0 / request 4
exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7
Measured v1.0 82d964e0 | raw events, no phase grouping / clock shift; bars <0.02 ms use visibility floor; dotted forward slots are equal-width guides, NOT measured boundaries
target_rank0
draft
DS: 6 rounds
PS: 3 rounds
1
2
3
4
5
6
1
2
3
0
5
10
15
20
25
time (ms from step start, host monotonic clock)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
p2_prepare
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
verify_accept_prep
verify_sample_accept
verify_setup

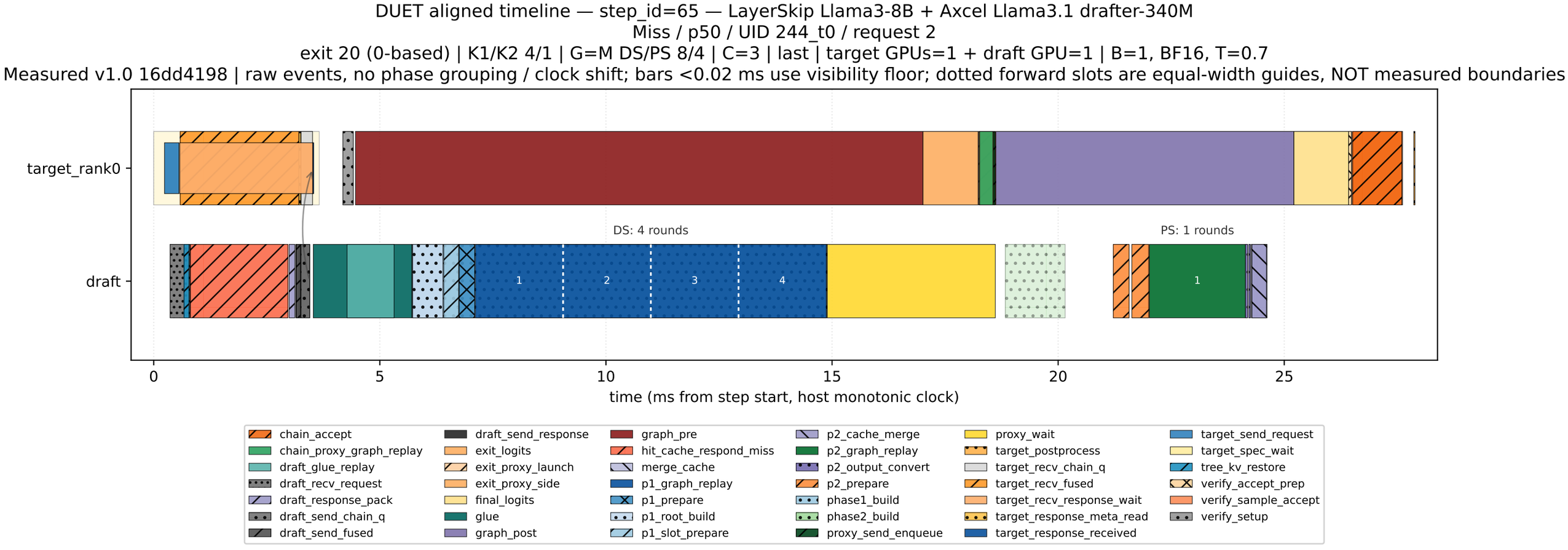

DUET aligned timeline — step_id=68 — LayerSkip Llama2-13B + AMD-Llama-135M
P1 hit / p50 / UID 84_t0 / request 1
exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7
Measured v1.0 59f46511 | raw events, no phase grouping / clock shift; bars <0.02 ms use visibility floor; dotted forward slots are equal-width guides, NOT measured boundaries
target_rank0
draft
DS: 6 rounds
PS: 3 rounds
1
2
3
4
5
6
1
2
3
0
5
10
15
20
25
30
35
time (ms from step start, host monotonic clock)
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p1
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
p2_prepare
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p1
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_parent_q_gather_p1
tree_hit_rerank_p1
tree_kv_commit
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

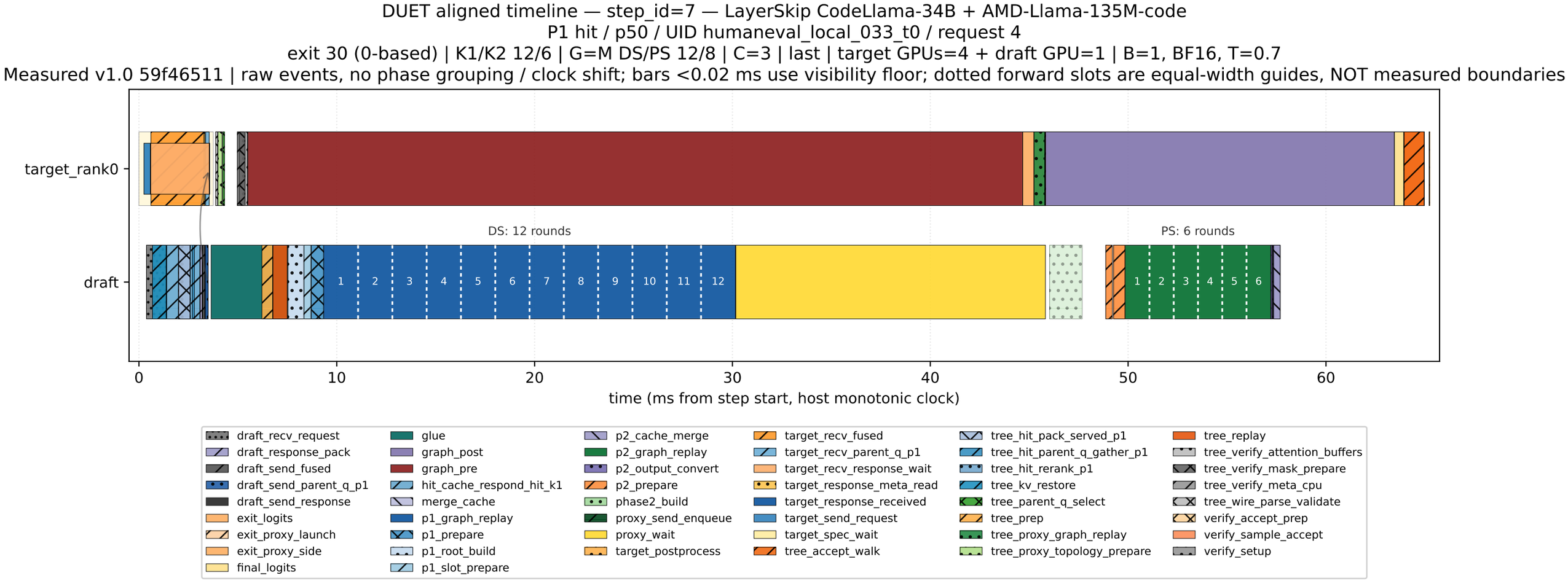

DUET aligned timeline — step_id=127 — LayerSkip Llama2-13B + AMD-Llama-135M
P2 hit / p50 / UID 288_t0 / request 4
exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7
Measured v1.0 59f46511 | raw events, no phase grouping / clock shift; bars <0.02 ms use visibility floor; dotted forward slots are equal-width guides, NOT measured boundaries
target_rank0
draft
DS: 6 rounds
PS: 3 rounds
1
2
3
4
5
6
0
5
10
15
20
25
30
time (ms from step start, host monotonic clock)
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k2
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
p2_prepare
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p2
tree_hit_rerank_p2
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

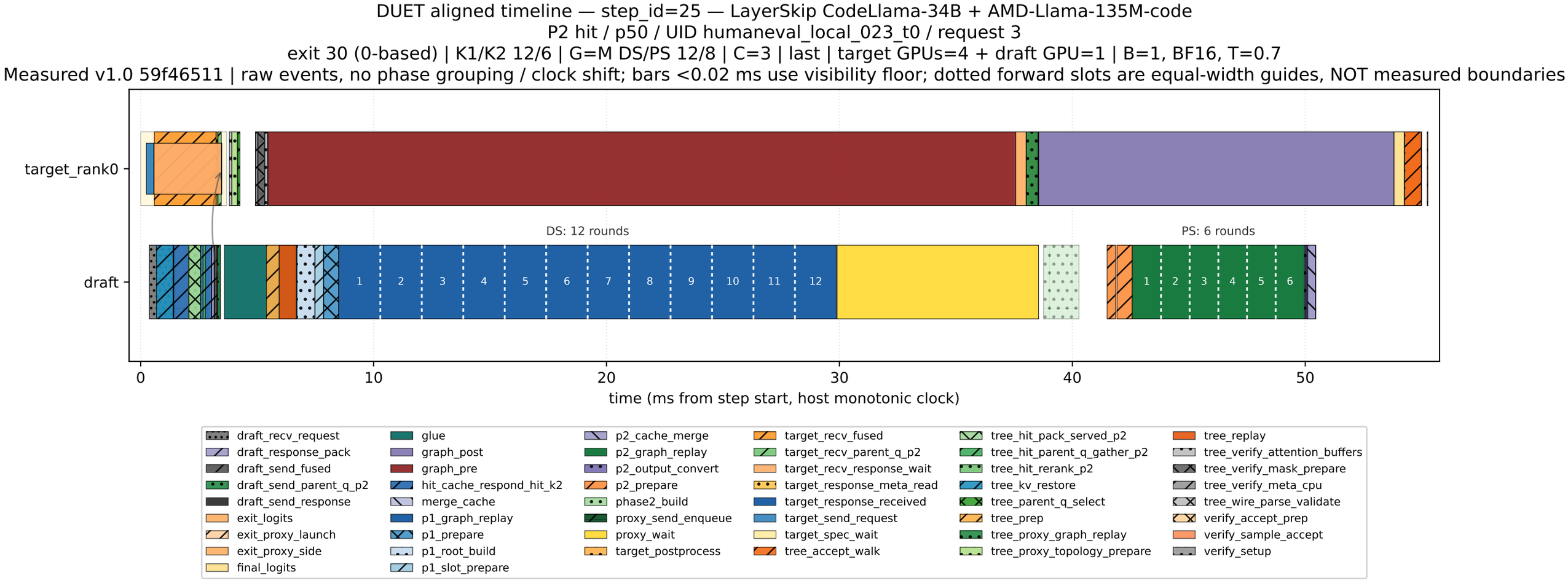

DUET aligned timeline — step_id=29 — LayerSkip Llama2-13B + AMD-Llama-135M
Miss / p50 / UID 128_t0 / request 3
exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7
Measured v1.0 59f46511 | raw events, no phase grouping / clock shift; bars <0.02 ms use visibility floor; dotted forward slots are equal-width guides, NOT measured boundaries
target_rank0
draft
DS: 6 rounds
PS: 3 rounds
1
2
3
4
5
6
1
2
3
0
5
10
15
20
25
30
time (ms from step start, host monotonic clock)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
p2_prepare
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_kv_restore
verify_accept_prep
verify_sample_accept
verify_setup

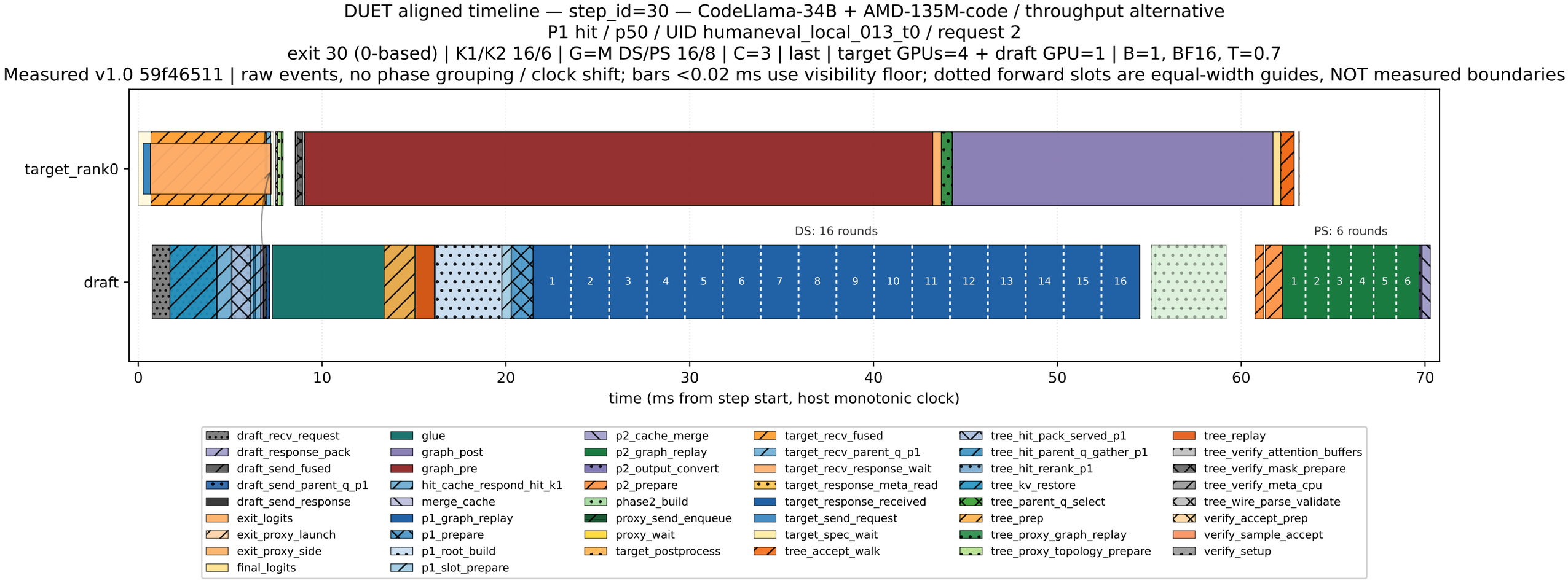

DUET aligned timeline — step_id=170 — LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M
P1 hit / p50 / UID 84_t0 / request 1
exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7
Measured v1.0 16dd4198 | raw events, no phase grouping / clock shift; bars <0.02 ms use visibility floor; dotted forward slots are equal-width guides, NOT measured boundaries
target_rank0
draft
DS: 4 rounds
PS: 1 rounds
1
2
3
4
1
0
5
10
15
20
25
time (ms from step start, host monotonic clock)
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p1
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
p2_prepare
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p1
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_parent_q_gather_p1
tree_hit_rerank_p1
tree_kv_commit
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

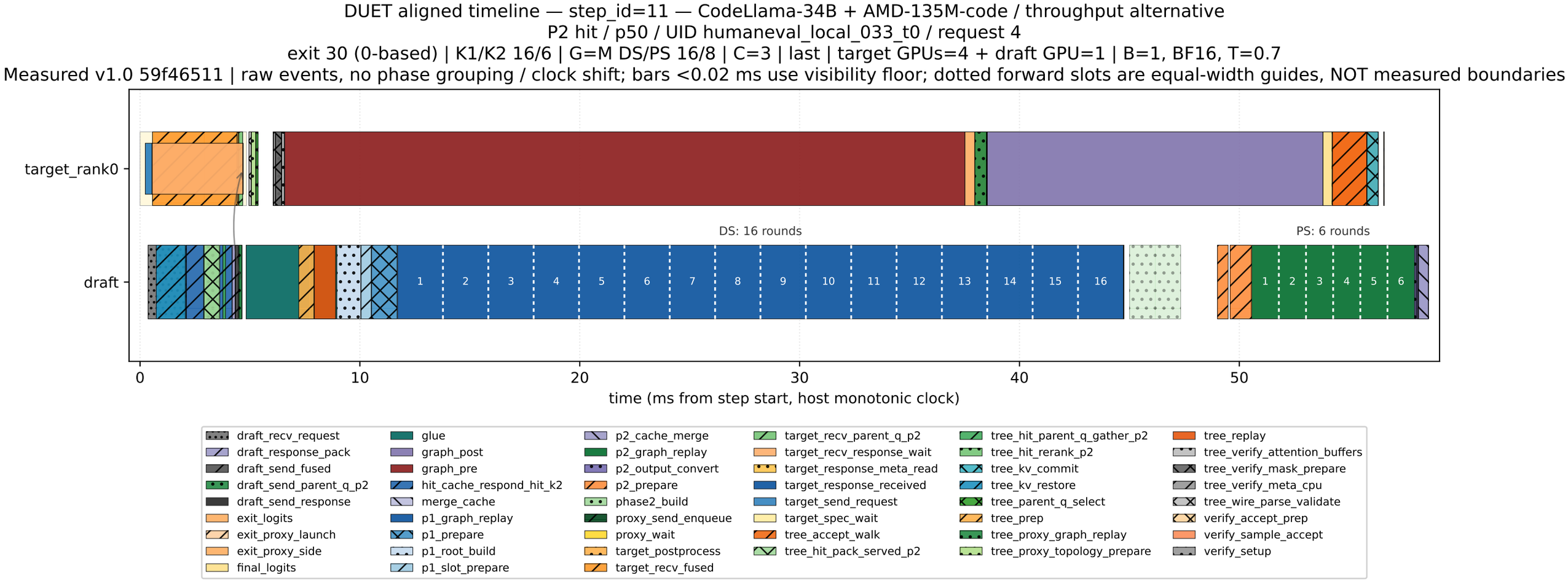

DUET aligned timeline — step_id=92 — LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M
P2 hit / p50 / UID 244_t0 / request 2
exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7
Measured v1.0 16dd4198 | raw events, no phase grouping / clock shift; bars <0.02 ms use visibility floor; dotted forward slots are equal-width guides, NOT measured boundaries
target_rank0
draft
DS: 4 rounds
PS: 1 rounds
1
2
3
4
1
0
5
10
15
20
25
time (ms from step start, host monotonic clock)
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k2
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
p2_prepare
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p2
tree_hit_rerank_p2
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

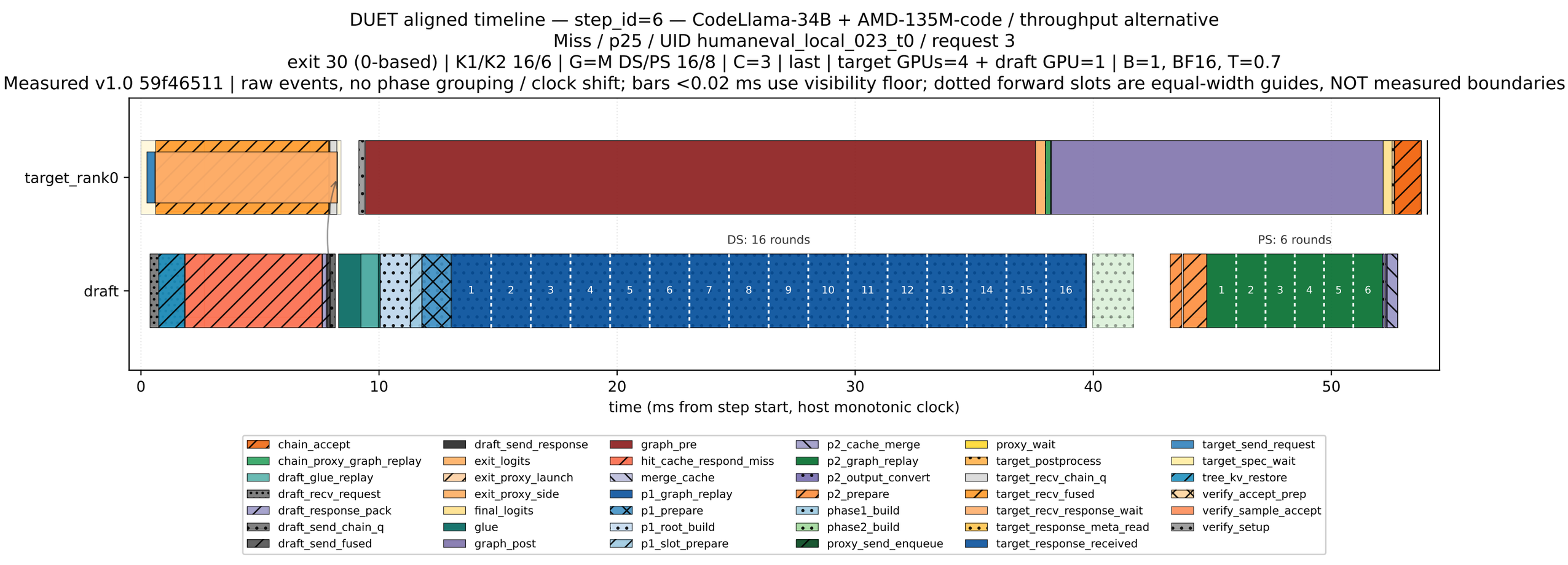

DUET aligned timeline — step_id=65 — LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M
Miss / p50 / UID 244_t0 / request 2
exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7
Measured v1.0 16dd4198 | raw events, no phase grouping / clock shift; bars <0.02 ms use visibility floor; dotted forward slots are equal-width guides, NOT measured boundaries
target_rank0
draft
DS: 4 rounds
PS: 1 rounds
1
2
3
4
1
0
5
10
15
20
25
time (ms from step start, host monotonic clock)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
p2_prepare
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_kv_restore
verify_accept_prep
verify_sample_accept
verify_setup


DUET aligned timeline — step_id=7 — LayerSkip CodeLlama-34B + AMD-Llama-135M-code
P1 hit / p50 / UID humaneval_local_033_t0 / request 4
exit 30 (0-based) | K1/K2 12/6 | G=M DS/PS 12/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7
Measured v1.0 59f46511 | raw events, no phase grouping / clock shift; bars <0.02 ms use visibility floor; dotted forward slots are equal-width guides, NOT measured boundaries
target_rank0
draft
DS: 12 rounds
PS: 6 rounds
1 2 3 4 5 6 7 8 9 10 11 12
1 2 3 4 5 6
0 10 20 30 40 50 60
time (ms from step start, host monotonic clock)
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p1
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
p2_prepare
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p1
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_parent_q_gather_p1
tree_hit_rerank_p1
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup


DUET aligned timeline — step_id=25 — LayerSkip CodeLlama-34B + AMD-Llama-135M-code
P2 hit / p50 / UID humaneval_local_023_t0 / request 3
exit 30 (0-based) | K1/K2 12/6 | G=M DS/PS 12/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7
Measured v1.0 59f46511 | raw events, no phase grouping / clock shift; bars <0.02 ms use visibility floor; dotted forward slots are equal-width guides, NOT measured boundaries
target_rank0
draft
DS: 12 rounds
PS: 6 rounds
1 2 3 4 5 6 7 8 9 10 11 12
1 2 3 4 5 6
0 10 20 30 40 50
time (ms from step start, host monotonic clock)
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k2
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
p2_prepare
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p2
tree_hit_rerank_p2
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup


DUET aligned timeline — step_id=30 — CodeLlama-34B + AMD-135M-code / throughput alternative
P1 hit / p50 / UID humaneval_local_013_t0 / request 2
exit 30 (0-based) | K1/K2 16/6 | G=M DS/PS 16/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7
Measured v1.0 59f46511 | raw events, no phase grouping / clock shift; bars <0.02 ms use visibility floor; dotted forward slots are equal-width guides, NOT measured boundaries
target_rank0
draft
DS: 16 rounds
PS: 6 rounds
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
1 2 3 4 5 6
0 10 20 30 40 50 60 70
time (ms from step start, host monotonic clock)
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p1
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
p2_prepare
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p1
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_parent_q_gather_p1
tree_hit_rerank_p1
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup


DUET aligned timeline — step_id=11 — CodeLlama-34B + AMD-135M-code / throughput alternative
P2 hit / p50 / UID humaneval_local_033_t0 / request 4
exit 30 (0-based) | K1/K2 16/6 | G=M DS/PS 16/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7
Measured v1.0 59f46511 | raw events, no phase grouping / clock shift; bars <0.02 ms use visibility floor; dotted forward slots are equal-width guides, NOT measured boundaries
target_rank0
draft
DS: 16 rounds
PS: 6 rounds
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
1 2 3 4 5 6
0
10
20
30
40
50
time (ms from step start, host monotonic clock)
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k2
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
p2_prepare
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p2
tree_hit_rerank_p2
tree_kv_commit
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup


DUET aligned timeline — step_id=6 — CodeLlama-34B + AMD-135M-code / throughput alternative
Miss / p25 / UID humaneval_local_023_t0 / request 3
exit 30 (0-based) | K1/K2 16/6 | G=M DS/PS 16/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7
Measured v1.0 59f46511 | raw events, no phase grouping / clock shift; bars <0.02 ms use visibility floor; dotted forward slots are equal-width guides, NOT measured boundaries
target_rank0
draft
DS: 16 rounds
PS: 6 rounds
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
1 2 3 4 5 6
0 10 20 30 40 50
time (ms from step start, host monotonic clock)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
p2_prepare
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_kv_restore
verify_accept_prep
verify_sample_accept
verify_setup